

# Replacing the Banker with a Function

## A Mathematician's Guide to Decentralized Finance

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# Outline

- 1 I. The Global Picture — Architecture & Vision
- 2 II. Market Mechanics — How DeFi Works
- 3 III. Synthesis — Mechanism Design & The Future

## I. The Global Picture — Architecture & Vision

*“Trust is not a leap of faith; it is the deterministic output of a distributed state machine.”*

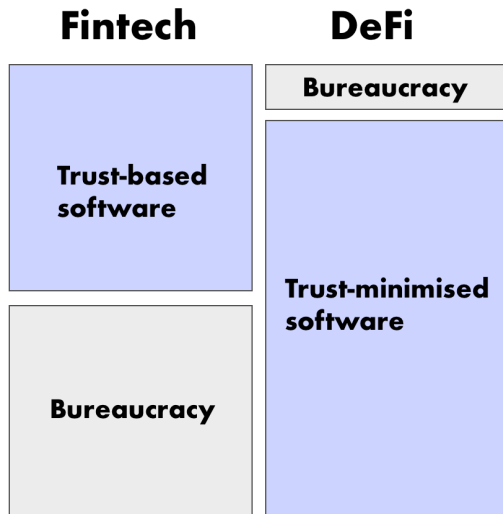
# Defining the Transformation: Fintech vs. DeFi

Before we dive into the math, we must distinguish between *digitalising* a bank and *rebuilding* finance.

- **Fintech (Digitised TradFi):** Centralised institutions using modern tech stacks. You have an app, but the bank still holds the *Master Ledger* and the *Kill Switch*.
- **DeFi (Decentralised Finance):** A peer-to-peer ecosystem where *Smart Contracts* act as the custodian and clearinghouse.
- **The Paradigm Shift:** Moving from *Subjective Trust* (Law/Reputation) to *Objective Truth* (Cryptographic Proofs).

References: [What is DeFi?](#) | [DeFi Beyond the Hype \(Wharton\)](#) | [CCAF DeFi Ecosystem Map](#)

# Defining the Transformation: Fintech vs. DeFi



# The Blockchain as a Global State Machine

For a mathematician, a blockchain is simply a **distributed, deterministic transition system**.

- **The State ( $S$ ):** A snapshot of all balances and contract data at time  $t$ .
- **The Transition Function ( $f$ ):** The rules of the protocol (the code).
- **The Formula:**

$$S_{t+1} = f(S_t, \Delta)$$

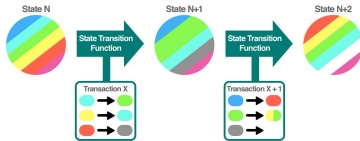
where  $\Delta$  represents a block of transactions.

- **Integrity:** Thousands of nodes must compute the same  $S_{t+1}$  and reach consensus.

**References:** [What is a Blockchain?](#) | [Ethereum & Bitcoin Tx Visualiser](#) | [Dan Boneh: Blockchain Primitives](#)

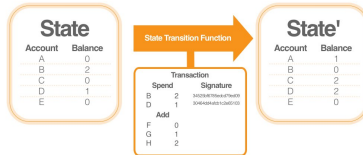
## Blockchain Computer State Systems

### State Transition Machine



$\text{State\_N+1} = \text{state\_transition\_function}(\text{State\_N}, \text{Transaction\_X})$

### Bitcoin State Transition System



**State** - ownership status of all existing BTC; accounts & BTC balances

**Transaction** - made of inputs (unspent BTC; sending address, signature) and outputs (receiving address, amount)

**State Transition Function** - applies a transaction to the current state, capable of account balance addition & subtraction

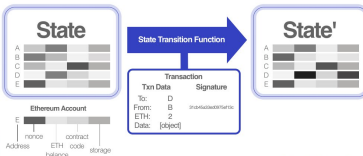
### The Universal State Machine

A **Turing Machine** is a mathematical model of computation used to investigate the theoretical limits of computer science. Using a small set of simple rules, the machine is capable of simulating ANY computer algorithm.



Turing Complete = system can solve any computational problem

### Ethereum State Transition System



Twitter: @SalomonCrypto

**State** - current status of all Ethereum accounts. An account can be controlled by private keys or contract code

**Transaction** - contains core cryptocurrency functionality plus an optional data field, accessible by smart contracts

**State Transition Function** - applies a transaction to the current state, processing the EVM, capable of processing arbitrary code

# The Engine of DeFi: Ethereum & The EVM

To turn a ledger into a financial system, we need a **Universal Computer** capable of executing arbitrary logic.

- **Smart Contracts:** Self-executing programs whose terms are written directly into code. They are *autonomous* (no middleman) and *immutable* (cannot be changed once deployed).
- **The Ethereum Virtual Machine (EVM):** A global, decentralised CPU ensuring every node executes the same code and arrives at the same result—guaranteeing *mathematical determinism*.
- **Gas & the Halting Problem:** Every operation costs “Gas”—a resource allocation mechanism and a formal constraint on computation in a Turing-complete system.

References: [What is Ethereum?](#) | [What Are Smart Contracts?](#)

# The Engine of DeFi: Ethereum & The EVM

## Traditional Contracts



## Smart Contracts



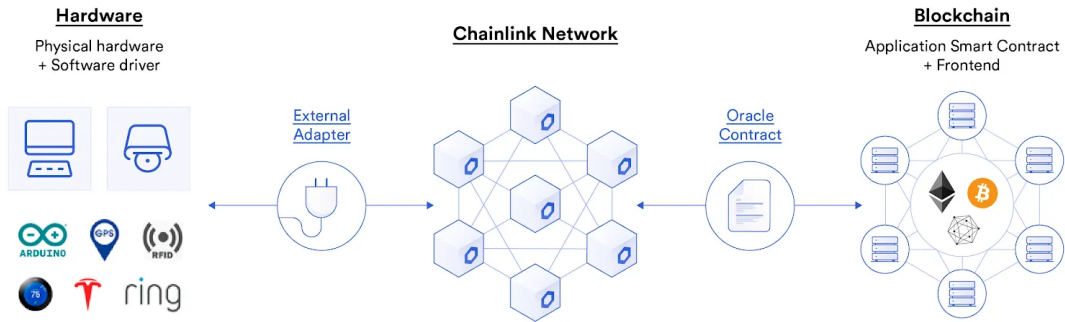
# Oracles: The Bridge to Reality

A closed mathematical system cannot natively “know” the price of Gold or the outcome of a game.

- **The Oracle Problem:** How do we bring external data into a deterministic system without introducing a single point of failure?
- **Decentralised Oracles:** Systems like *Chainlink* use a committee of nodes to provide a “consensus” data feed.
- **Application:** A smart contract can trigger a *liquidation* or a payout based on real-world events (e.g. ETH price dropping below \$1800).

References: [What Is a Blockchain Oracle?](#) | [What Is Chainlink?](#)

# Oracles: The Bridge to Reality



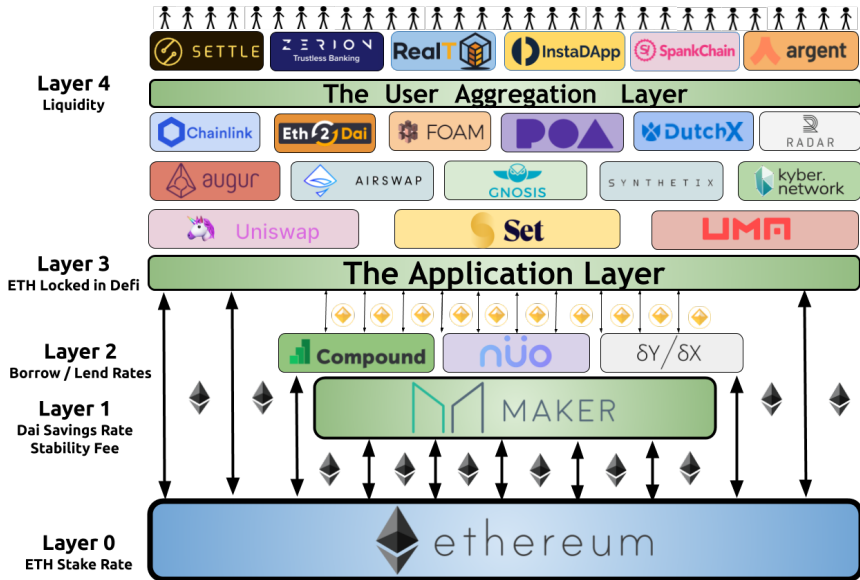
## Summary: The Architecture of Integrity

| Feature          | Traditional System      | DeFi System                        |
|------------------|-------------------------|------------------------------------|
| Enforcement      | Legal Prose (Ambiguous) | Mathematical Code (Precise)        |
| Custodian        | Centralised Bank        | Self-Custody / Smart Contract      |
| Verification     | Post-hoc Audits         | Real-time Cryptographic Proofs     |
| Interoperability | Siloed APIs             | Open Composability (“Money Legos”) |

### The “Money Lego” Framework

DeFi protocols are modular by design. Layer 1 provides settlement, tokens serve as raw materials, and lending/trading protocols act as building blocks. Developers “snap” these bricks together to create complex strategies without seeking institutional permission.

# Summary: The Architecture of Integrity



## II. Market Mechanics — How DeFi Works

*“Liquidity is an invariant, and price is the slope of a curve defined by the balance of reserves.”*

# Beyond Order Books: Automated Market Makers (AMMs)

In traditional markets, a trade requires a counterparty (Buyer  $\leftrightarrow$  Seller). In DeFi, you trade against a **Liquidity Pool**.

- **The Liquidity Pool:** A smart contract holding a pair of tokens (e.g.  $X$  and  $Y$ ).
- **The Constant Product Invariant:**

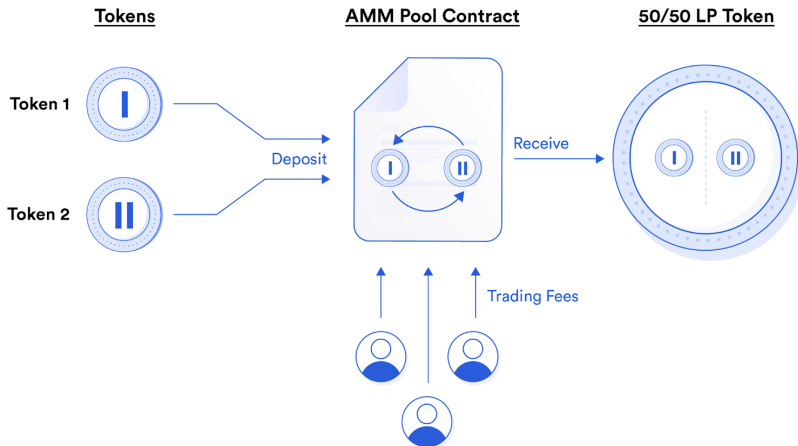
$$x \cdot y = k$$

where  $x$ ,  $y$  are token reserves and  $k$  is a constant maintained during a trade.

- **The “Peer-to-Pool” Model:** Anyone can become a *Liquidity Provider (LP)* by depositing assets, earning a pro-rata share of trading fees.

References: [What Are AMMs?](#) | [Mastering AMMs \(Three Sigma\)](#)

# Beyond Order Books: Automated Market Makers (AMMs)



# The Geometry of Price Discovery

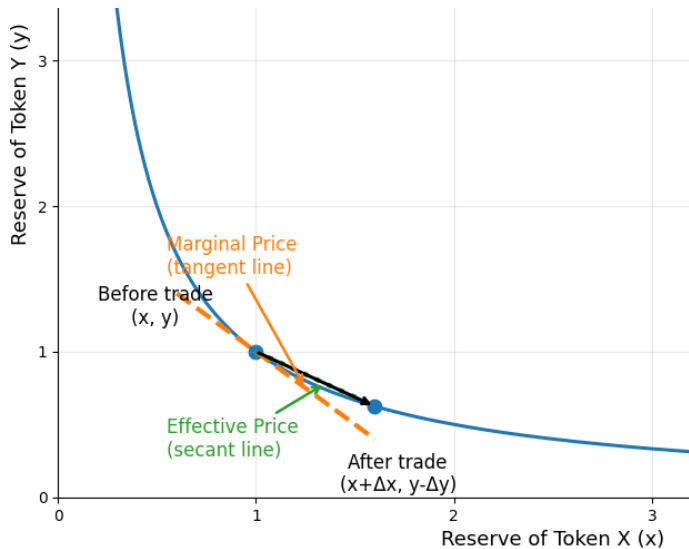
Price is not “set” by an admin; it is a **mathematical consequence** of the reserve ratio.

- **Marginal Price:** For  $y = k/x$ ,

$$P = \frac{dy}{dx} = -\frac{y}{x}$$

- **Slippage:** Because the curve is a hyperbola, every trade shifts  $y/x$ .
  - *Infinitesimal trades:* Price stays near spot  $y/x$ .
  - *Large trades:* Significant displacement along the curve.
- **Asymptotic Liquidity:** The pool can never be fully depleted—as quantity  $\rightarrow 0$ , price  $\rightarrow \infty$ .

# The Geometry of Price Discovery



## Worked Example: Trading on the Curve

### Setup

Pool holds 100 ETH ( $x$ ) and 200,000 USDC ( $y$ ).

$k = 100 \times 200,000 = 20,000,000$ . Spot price  $P = y/x = 2,000$  per ETH.

**Trade:** A trader wants to buy 10 ETH.

- New ETH reserve:  $x' = 100 - 10 = 90$ .
- New USDC reserve:  $y' = 20,000,000 / 90 = 222,222.22$ .
- Cost:  $222,222.22 - 200,000 = 22,222.22$  USDC.

### Result

Effective price: \$2,222.22 per ETH  $\implies \approx 11.1\%$  slippage from spot.

Protocols like **Aave** replace loan officers with a *Utilisation-based Interest Rate Model*.

- **Utilisation Rate:**

$$U = \frac{\text{Total Borrows}}{\text{Total Liquidity}}$$

- **Equilibrium Logic:**

- Low  $U$ : Excess supply  $\Rightarrow$  low rates to incentivise borrowing.
- High  $U$ : Scarcity  $\Rightarrow$  sharp spikes to attract lenders and force debt repayment.

# The Kinked Interest Rate Model

The rate function  $R(U)$  is piecewise linear around an *Optimal Utilisation*  $U_{\text{opt}}$ :

Healthy Zone ( $U < U_{\text{opt}}$ )

$$R_t = R_0 + \left( \frac{U_t}{U_{\text{opt}}} \right) R_{\text{slope1}}$$

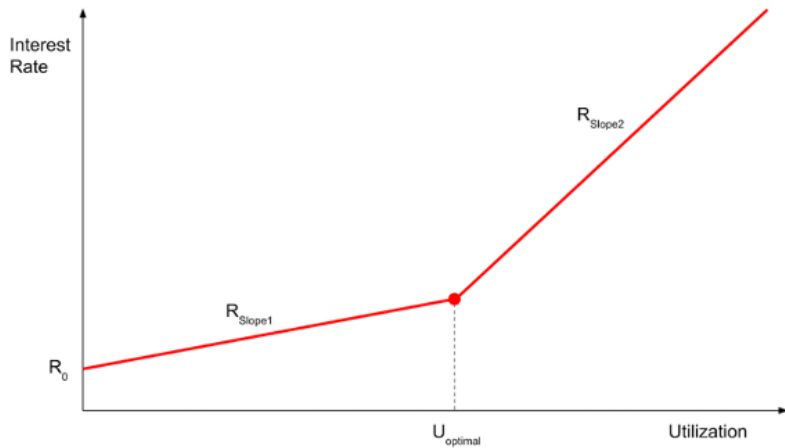
Emergency Zone ( $U \geq U_{\text{opt}}$ )

$$R_t = R_0 + R_{\text{slope1}} + \left( \frac{U_t - U_{\text{opt}}}{1 - U_{\text{opt}}} \right) R_{\text{slope2}}$$

**The “Kink”:**  $R_{\text{slope2}}$  is usually an order of magnitude higher than  $R_{\text{slope1}}$  (e.g. jumping from 4% to 100%)—a mathematical “emergency brake” to prevent pool depletion.

References: [DeFi Money Markets 2024](#) | [Aave Docs](#) | [Compound Docs](#)

# The Kinked Interest Rate Model



In a pseudonymous system without legal recourse, all credit is secured by a **Collateral Invariant**.

- **Collateralisation Ratio:**

$$CR = \frac{\sum(\text{Collateral Assets} \times \text{Price})}{\text{Total Debt Value}}$$

- **Loan-to-Value (LTV):** Maximum leverage allowed (e.g.  $LTV = 75\%$  implies a minimum  $CR$  of  $133\%$ ).
- **Oracle Dependency:** System safety is strictly bound to the accuracy and latency of external price feeds.

# The Liquidation Function

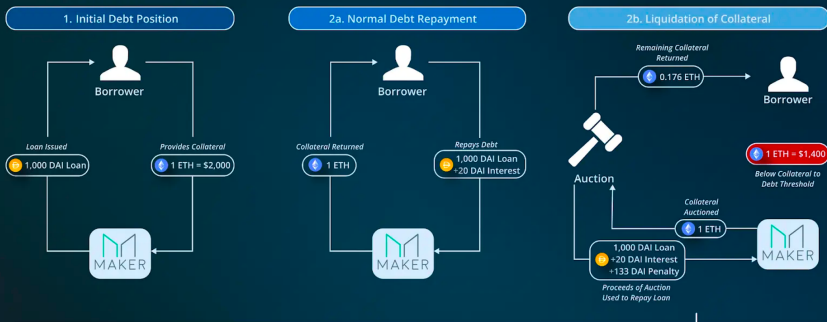
The “fail-safe” mechanism that maintains system-wide solvency.

- **Trigger:** If  $CR < \text{Liquidation Threshold}$ , the position is flagged as under-collateralised.
- **Forced Exit:** Automated “Liquidator” bots repay the debt in exchange for the collateral at a *Liquidation Bonus* (e.g. 5%–10% discount).
- **Mathematical Solvency:** By liquidating before  $CR < 100\%$ , the protocol ensures that every dollar of debt is always backed by more than one dollar of market value.



## Simplified Overview of the Maker Protocol

Under normal circumstances, DAI loans are repaid in full plus interest but, vaults may be liquidated if the collateral to debt ratio falls below the threshold



## Liquidation Walkthrough: A “Hard Money” Event

- 1 **Open Position:** Deposit 1 ETH (\$2,000). Borrow 1,500 USDC ( $LTV = 75\%$ ).
- 2 **Current State:**  $CR = 2,000/1,500 = 133.3\%$  (Healthy).
- 3 **Market Shock:** ETH drops to \$1,700.
- 4 **Breach:**  $CR = 113.3\%$ . Threshold is  $120\% \Rightarrow$  position is now **liquidatable**.
- 5 **Execution:** A bot detects the breach, repays 1,500 USDC, and claims the ETH at a discount.

### Takeaway

Debt is cleared by code, not by credit committees or bailouts. The math is the ultimate arbiter of solvency.

References: [What Are Stablecoins?](#) | [Sky Protocol: Multi-Collateral Dai](#)

### III. Synthesis — Mechanism Design & The Future

*“The bedrock of future finance is not the institution,  
but the formally verified proof.”*

# Mechanism Design: Engineering Rational Behaviour

In a permissionless system, we cannot “ban” bad actors; we must make it **mathematically unprofitable** to be one.

- **Agent-Based Economics:** Model the protocol as a game where autonomous agents maximise their own utility.
- **Slashing & Incentives:**
  - *Positive:* Yield and rewards for honest participation.
  - *Negative:* Malicious behaviour results in automated forfeiture of staked capital (*Slashing*).
- **Sybil Resistance:** Proof of Stake imposes an economic cost that makes it prohibitively expensive to subvert the system via multiple identities.

References: [8 Reasons Why Blockchain Mechanism Design Is Hard](#) | [Permissionless Mechanism Design](#)

# When the Math Breaks: The Terra/Luna Collapse

In May 2022, the Terra ecosystem provided a \$40 billion case study in **mechanism failure**.

- **The Setup:** TerraUSD (UST) was an *algorithmic stablecoin* backed by a game-theoretic arbitrage loop with LUNA.
- **The Death Spiral:**

$UST < \$1 \Rightarrow \text{burn UST, mint LUNA} \Rightarrow \text{LUNA hyper-inflates} \Rightarrow \text{UST loses backing}$

- **The Lesson:** The system was *locally stable* but *globally fragile*. It assumed a constant presence of rational arbitrageurs and failed to account for “bank run” dynamics.

Reference: [Luna Brothers, Inc. \(BitMEX\)](#)

# Privacy via Zero-Knowledge Proofs (ZKP)

**The Transparency Paradox:** Blockchains are public, but finance requires privacy.

- **ZKP Definition:** A cryptographic method where a *Prover* convinces a *Verifier* that a statement is true (e.g. “I am solvent”) without revealing the underlying data (e.g. “my balance”).
- **Mathematical Applications:**
  - *Private Credit:* Prove Collateral  $>$  Debt without revealing wallet history.
  - *Identity:* Prove you are verified or  $>18$  without sharing your name.
  - *Scalability (zk-Rollups):* Bundle 10,000 transactions into a single succinct proof verifiable in milliseconds.

**References:** [ZKP: An Illustrated Primer](#) | [Decentralised Identity Guide 2025](#)

# Open Problems: The Frontiers of DeFi

An invitation for students to apply advanced mathematics to unsolved challenges:

- **Formal Verification:** Using formal methods (e.g.  $\text{TLA}^+$ , Coq) to *prove* a smart contract is bug-free before it handles billions of dollars.
- **Systemic Risk & Contagion:** Modelling the “Money Lego” effect—how the failure of one protocol propagates through the hyper-connected liquidity network.
- **MEV (Maximal Extractable Value):** Studying the game theory of transaction ordering to prevent sophisticated bots from front-running users.

# Your Career: Beyond the Classroom

Digital Finance is a rare field where **pure math translates directly into production infrastructure**.

- **Protocol Architect:** Designing the next generation of invariant curves and interest rate models.
- **Risk Quantitative (Quant):** Running Monte Carlo simulations for firms like [Gauntlet](#) or [Chaos Labs](#) to stress-test protocol parameters.
- **The Opportunity:** You are at the intersection of *Cryptography*, *Game Theory*, and *Quantitative Finance*. The “Wild West” era is ending; the *Era of Rigor* is beginning.

## Final Closing Thought

*“Digital Finance is not about ‘disrupting’ banks; it is about building a financial system that is as predictable and transparent as the laws of mathematics.”*

Thank you!



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